

COL761 Assignment 2: Q1 Report

K-Means Model Selection

1 Introduction

Clustering is an unsupervised learning method used to group similar data points together. In this assignment, k-means clustering was applied to the given datasets. The goal was to determine a suitable number of clusters by evaluating the k-means objective for values of k from 1 to 15 and analyzing the resulting elbow plots.

2 K-Means Objective Function

For a dataset $\{x_1, x_2, \dots, x_n\}$ in $\mathbb{R}^d$, k-means minimizes:

$$J = \sum_{i=1}^n \|x_i - \mu_{c_i}\|^2$$

where x_i is a data point and μ_{c_i} is the centroid of the cluster assigned to x_i . This is also known as the **within-cluster sum of squares (WCSS)** or the **k-means objective**. Lower values indicate more compact clusters.

3 Methodology

For each dataset, k-means was run for all values of $k \in \{1, 2, \dots, 15\}$. For every value of k , the corresponding WCSS value was recorded and plotted against k . The elbow point was automatically detected using the maximum perpendicular distance from the line connecting the first and last points of the objective curve.

4 Empirical Results

Figure 1 shows the elbow plots for Dataset 1 and Dataset 2.

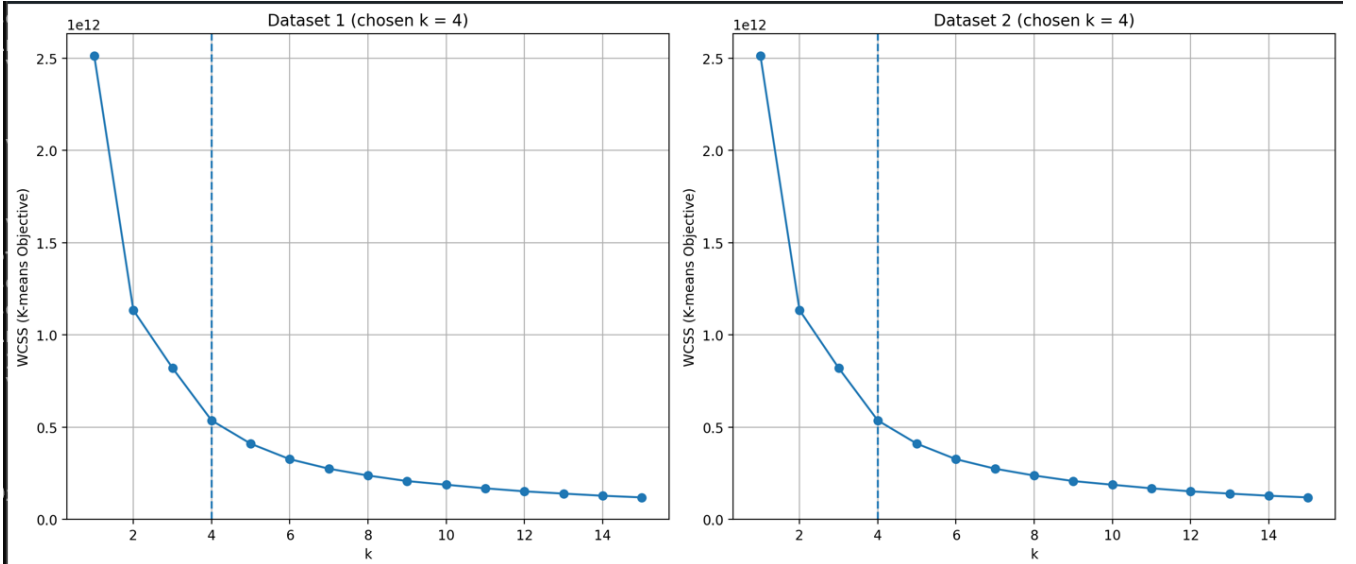


Figure 1: Elbow plots for Dataset 1 and Dataset 2.

From the plots, the objective decreases sharply for small values of k and then begins to flatten, forming an elbow in the curve. Using the maximum distance from the line joining the first and last points (a standard elbow detection heuristic), the selected number of clusters is $k = 4$ for both datasets.

5 Structural and Statistical Assumptions of K-Means

The effectiveness of k-means depends on several structural and statistical assumptions:

- **Spherical clusters:** Clusters are assumed to be approximately spherical in Euclidean space.
- **Equal variance:** The algorithm implicitly assumes that different clusters have similar variance.
- **Balanced cluster sizes:** Extremely imbalanced clusters can distort centroid estimates.
- **Euclidean distance appropriateness:** Euclidean distance is assumed to be a meaningful measure of similarity.

When these assumptions are reasonably satisfied, k-means tends to produce meaningful partitions of the data.

6 Limitations of the Model Selection Procedure

The elbow method used for selecting k has several limitations:

- The elbow may not always be clearly visible in the objective curve.
- Different random initializations can produce slightly different objective values.
- The method relies partly on visual inspection, which introduces subjectivity.
- It does not directly measure clustering quality beyond the optimization objective.

Thus, the elbow method should be viewed as a useful heuristic rather than a definitive statistical criterion.

7 Possible Methodological Improvements

The model selection procedure can be strengthened using alternative validation approaches:

- **Silhouette Score:** Evaluates how well each point fits its own cluster relative to other clusters.
- **Gap Statistic:** Compares clustering performance against a random reference distribution.
- **Information Criteria:** Measures such as BIC or AIC can be used in probabilistic clustering settings.
- **Stability Analysis:** Examines whether cluster assignments remain consistent across resampled datasets.

These methods provide more principled statistical validation than the basic elbow heuristic.

8 Failure Modes of K-Means

K-means may perform poorly in several common scenarios:

- **Non-spherical clusters:** When the true clusters are elongated or irregular.
- **Different densities:** Dense clusters may be split while sparse clusters may be merged.
- **Presence of outliers:** Outliers can significantly shift the centroids.
- **Non-linear cluster boundaries:** When separation requires non-linear decision boundaries.

In such cases, alternative methods such as DBSCAN, spectral clustering, or Gaussian mixture models may be more suitable.

9 Conclusion

In this study, the k-means objective was evaluated for values of k from 1 to 15 for the given datasets. The elbow plots show a clear bend around $k = 4$ for both Dataset 1 and Dataset 2, indicating that additional clusters beyond this point provide only limited improvement. Hence, $k = 4$ was selected as the appropriate number of clusters for both datasets. Although the elbow method is intuitive and easy to use, its limitations suggest that complementary validation techniques may be helpful in more complex clustering problems.